Python Dictionaries to Pandas DataFrames

Data with Dylan | Cheat Sheet | Python Fundamentals #7

WHAT IS A DATAFRAME?

A DataFrame is Pandas' core 2D table.
It stores, views, and analyzes data.

💡 Think of it as a spreadsheet: columns are fields; each row is one record.

TYPE / FORMATTING WARNINGS

▲ **Match list lengths**
Each dictionary value list needs the same number of items.

▲ **Use exact casing**
Write pd.DataFrame(...), not pd.dataframe(...) or dataframe.

▲ **Import first**
Run import pandas as pd before calling pd.DataFrame().

KEY IDEA: HOW ROWS FORM

```
# first item from every list -> Row 0
# second item from every list -> Row 1
# keys become the column headers
```

⚡ MINI PRACTICE

df already exists.
Write one line that exports it as college_data.csv without saving the index column.

Hints: use .to_csv(...) • include index=False

DICTIONARY -> DATAFRAME

DICTIONARY COMPONENT	DATAFRAME COMPONENT	VISUAL LAYOUT DESCRIPTION
`key`	column header	"University" becomes a column name.
`value list`	data column	List items fill one row at a time.
`same list index`	row	Matching list positions form Row 0.

KEY SYNTAX & OPERATIONS

1. Import Pandas

```
import pandas as pd
```

2. Create the dictionary

```
college_data = {"University": ["UW", "WSU"],
                "Tuition": [12600, 12100]}
```

3. Convert to a DataFrame

```
df = pd.DataFrame(college_data)
```

4. Inspect the result

```
df        # view the table
df.head() # first 5 rows
```

COMMON BEGINNER MISTAKES

X **Wrong import alias**
`import pd`
✓ `import pandas as pd`

X **Lowercase DataFrame class**
`pd.dataframe(college_data)`
✓ `pd.DataFrame(college_data)`

X **Forgetting parentheses**
`df.head`
✓ `df.head()`

X **Mismatched list lengths**
`["UW", "WSU"] vs [12600]`
✓ `use the same count per list`